

Testamentum Jacobi Balcerkovic

EML Project

Jude Lynch

Comparentes personaliter Honoratus Andreas Simonovici
Scabinus Civitatis Leopoldi locum Ordinarii Advocati
Leopoldi sub tempore pestifero hinc inde breviter tenens
et Famuli mei
et Matias Scabinorum
pro tempore illi
publice benevolens eorum relationem suam fidelem
fecerunt: Quia vestri
vici Dolitor Civis
infecus esset, mortis se propinquum animadvertens,
morbis se tolerare non posse videret postulavit per
certum nuncium ab ipsis, ut se adirent et ultimam vo-
luntatem suam exaudirent. Quam demum ubi opus
esset testificarentur. Adierunt itaque ex Officio nostro

3 Step Pipeline

`main.py` - HTR text extraction

`entity_graph.py` - Single page NER extraction

`entity_registry.py` - Multi-corpus Canonicalization

`graph_builder.py` - Cross page relation extractor, graph assembler

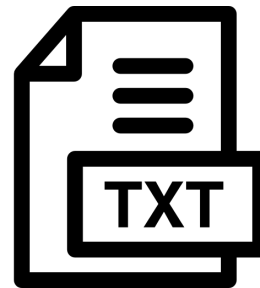
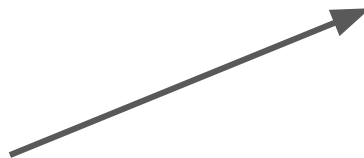
OCR Abstraction Process - main.py



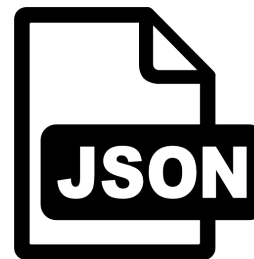
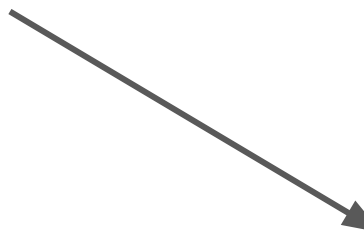
`main.py`



Book 23
scans



Raw output



OCR output
reasoning
(`PAGE.JPG.words.json`)

23-2-52 > 0005.JPG.parsed.txt

1 [LA]Actum Leopoli feria secunda post
2 festum Circumcisionis Domini pro-
3 xima Anno Domini Millesimo Quin-
4 gesimo nonagesimo quinto
5 Testamentum Jacobi [PL]Balcertowic[LA]
6 Comparentes personaliter Honoratus Andreas [PL]Simonowic[LA]
7 Scabinus Civitatis Leopoliensis locum Ordinarii Advocati
8 Leopoli sub tempore pestifero proximo praeterito tenens
9 et Famati Christopherus [PL]Mozanc[LA] Bartholomeus [PL]Strach[LA]
10 et Matias [PL]Sapiha[LA] Cives Leopolienses in locum Scabinorum
11 pro tempore illo pestifero absentium assumpti palam
12 publice benevolentque eiusmodi relationem suam fidelem
13 fecerunt: Quia vixit Famatus olim Jacobus [PL]Balcerto-
14 wic[LA] Doliator Civis Leopoliensis dum pestifera contagione
15 infectus esset, mortique se propinquum animadverteret,
16 morbumque se tolerare non posse videret postulavit per
17 certum nuntium ab ipsis, ut se adirent et ultimam vo-
18 luntatem suam exaudirent. Quam deinde ubi opus
19 esset testificarentur. Adierunt itaque ex Officio nostro
20 additi ultimamque eius voluntatem eiusmodi in scripto
21 porrectam palam benevolentque professam exaudierunt
22 in tenore tali.
23 TENOR TESTAMENTI.
24 [PL]Ja Jakub Balcerowic Bednarz bacząc sie być chorym y
25 zdrowia niebespiecznego, wszakże będąc baczością z łaski
26 Panskiej dobrze przed Urzędem w mscie dobrowolnie ze-
27 znawam. Iż gdzieby Pan Bog na me w tey niemocy moiej
28 śmierć dopuścić raczył, ia rzeczy ruchome moje wszystkie
29 żenie moiej Zofiey za iey uczciwe zachowanie od mała do
30 wiela żadnych nie wyjmując zapisuie, daruie i odkasuie
31 tym testamentem wiecznym darem nic w nich prawa Przyia-
32 ciołom moim i krewnym nie zostawiając, ktore rzeczy ona.

23-2-52 > {} 0005.JPG.words.json > {} words > {} 3 > # language_confidence_score

```
1 {  
2   "source_image": "0005.JPG",  
3   "page_number": 5,  
4   "words": [  
5     {  
6       "word": "Actum",  
7       "language": "Latin",  
8       "language_confidence_score": 1.0,  
9       "language_confidence_reasoning": "Standard Latin formulaic legal opening.",  
10      "word_declension": null,  
11      "word_type": "verb",  
12      "line_number": 1,  
13      "transcription_confidence_score": 1.0,  
14      "transcription_confidence_reasoning": "Clear calligraphic gothic script."  
15    },  
16    {  
17      "word": "Leopoli",  
18      "language": "Latin",  
19      "language_confidence_score": 1.0,  
20      "language_confidence_reasoning": "Latin locative form of Lviv.",  
21      "word_declension": null,  
22      "word_type": "location",  
23      "line_number": 1,  
24      "transcription_confidence_score": 1.0,  
25      "transcription_confidence_reasoning": "Clearly written."  
26    },  
27    {  
28      "word": "feria",  
29      "language": "Latin",  
30      "language_confidence_score": 1.0,  
31      "language_confidence_reasoning": "Standard Latin day designation.",  
32      "word_declension": null,  
33      "word_type": "other",  
34      "line_number": 1,  
35      "transcription_confidence_score": 1.0,  
36      "transcription_confidence_reasoning": "Clearly written."  
37    },  
38    {  
39      "word": "secunda",  
40      "language": "Latin",
```

Entity Graph Process - entity_graph.py

```
{
  "document_id": "23-2-52/0005.JPG.txt",
  "entities": [
    {
      "entity_id": "E1",
      "entity_type": "location",
      "canonical_name": "Lviv",
      "mention_texts": [
        "Leopoli",
        "Civitatis Leopoliensis",
        "Leopolienses",
        "mscie"
      ],
      "justification": "Actum Leopoli... Scabinus"
    },
  ],
}
```

```
{  
  "entity_id": "E3",  
  "entity_type": "person",  
  "canonical_name": "Andreas Simonowic",  
  "mention_texts": [  
    "Andreas Simonowic"  
  ],  
  "justification": "Honoratus Andreas Simonowic"  
},
```

```
  "relationships": [  
    {  
      "head_entity_id": "E3",  
      "relation": "holds_office_in",  
      "tail_entity_id": "E1",  
      "evidence": "Andreas Simonowic Scabinus Civit"  
    },  
  ],
```

Multi-Corpus Canonicalization - entity_graph.py

```
{  
  "entity_id": "LOC-0001",  
  "entity_type": "location",  
  "canonical_name": "Lviv",  
  "aliases": [  
    "Lviv",  
    "м.Львів",  
    "LEOPOLIENSI",  
    "LEOPOLIENSIS",  
    "Leopoli",  
    "Civitatis Leopoliensis",  
    "Leopolienses",  
    "mscie",  
    "Suburbani Leopolitani",  
    "Leopolitano",  
    "Civis Leopoliensis",  
    "Civi Leopoliensi",  
    "Suburbano Leopoliensi",  
    "Judicio Civili Leopoliensi",  
    "Leopoliensibus",  
    "Leopoliensem",  
    "Civium Leopoliensium",  
    "LEPOENSI"  
  ],  
  "source": "llm",  
  "merged_from": [],  
}
```

```
{  
  "document_id": "23-2-52/0007.JPG.txt",  
  "local_entity_id": "E6",  
  "mention_texts": [  
    "Leopoli",  
    "Suburbani Leopolitani",  
    "Leopolitano"  
  ],  
  "justification": "City where the court acts were issued and",  
  "extracted_at": "2026-09-14T06:09:09.892446+00:00"  
},  
{  
  "document_id": "23-2-52/0050.JPG.txt",  
  "local_entity_id": "E4",  
  "mention_texts": [  
    "Leopoli",  
    "Civis Leopoliensis",  
    "Civi Leopoliensi"  
  ],  
  "justification": "City where the legal acts take place and",  
  "extracted_at": "2026-09-14T06:09:22.069894+00:00"  
},
```

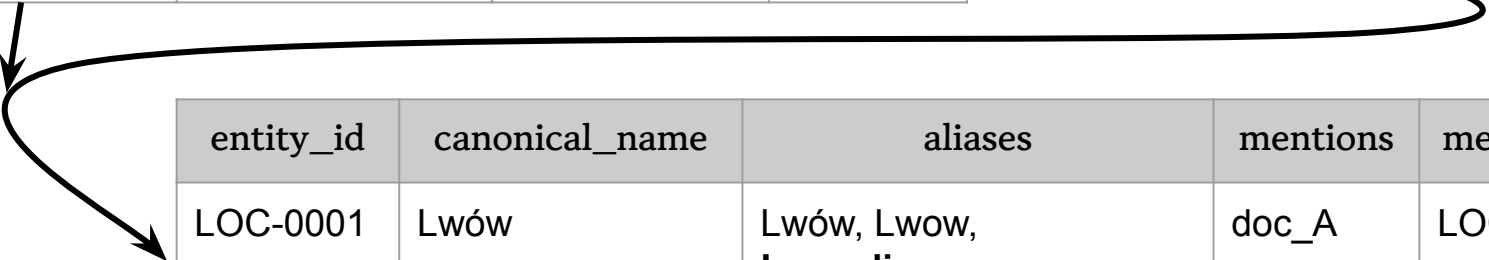
Manual Registry - Lookup

registry.json

entity_id	canonical_name	aliases	mentions
LOC-0001	Lwów	Lwów, Lwow	doc_A
LOC-0004	Leopolis	Leopolis	doc_B
PER-0002	Jan Kazimierz	Jan Kazimierz	doc_C

lookup.json

```
{  
  "entity_type": "location",  
  "canonical_name": "Lwów",  
  "aliases": ["Lviv", "Leopolis", "Львів"],  
  "notes": "English, Latin, German, and Ukrainian"  
},
```



entity_id	canonical_name	aliases	mentions	merged_from
LOC-0001	Lwów	Lwów, Lwow, Leopolis	doc_A	LOC-0004
PER-0002	Jan Kazimierz	Jan Kazimierz	doc_C	

Cross Relation Extractor - graph_builder.py



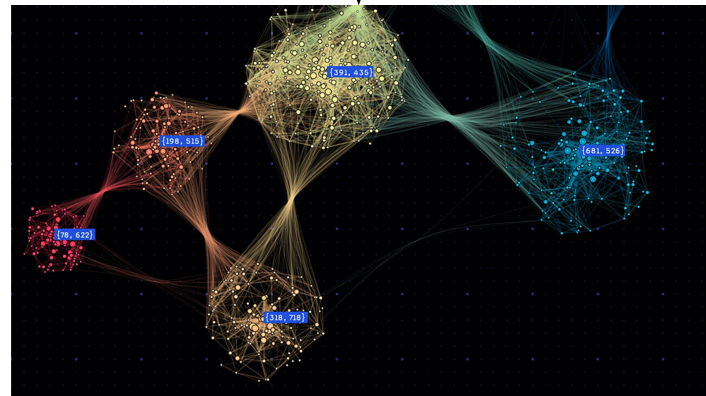
graph_builder.py

A screenshot of a terminal window showing a SQL query and its results. The query is "SELECT subject_id, predicate, object_id FROM triples". The results are displayed in a table with three columns: subject_id, predicate, and object_id. The table contains six rows of data.

	subject_id	predicate	object_id
20	PER-0017	reside_in	LOC-0001
21	PER-0018	reside_in	LOC-0001
22	PER-0018	marry	PER-0019
23	PER-0018	bequeath	PER-0019
24	PER-0018	appoint_executor	ORG-0003
25	PER-0018	bequeath	CUR-0001

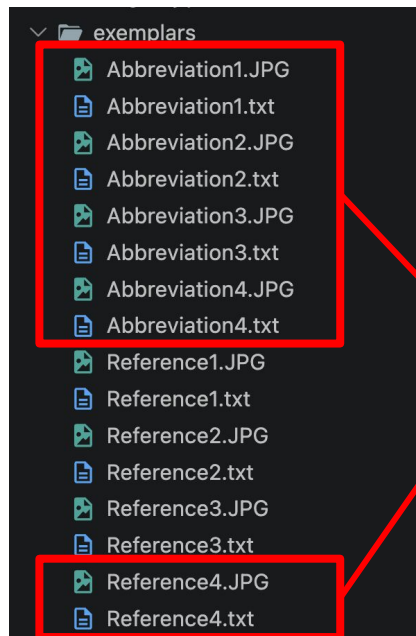


NetworkX



Current Limitations 1: Exemplars

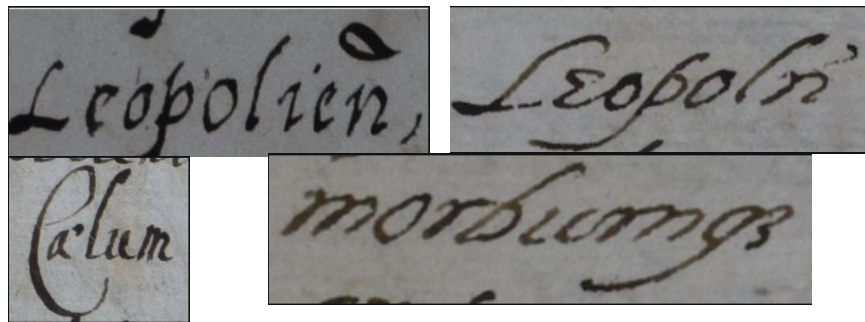
Exemplars in OCR extraction agent [main.py](#) are biased towards Latin. Limited scribal abbreviations with few books



Only Polish exemplar!

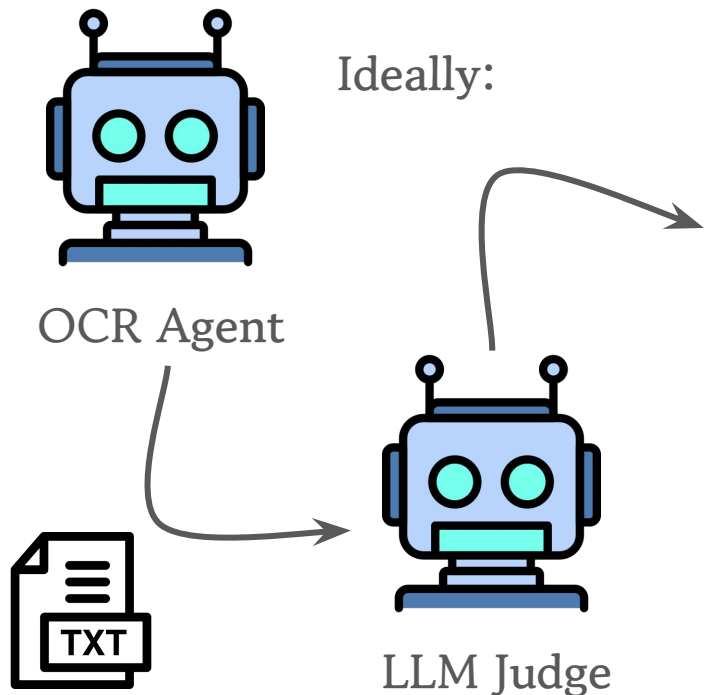
exemplars > Reference4.txt

```
1  Ja Jakub Balcerowit Bednarz bacząc sie byc chorym i
2  Zdrowia niebezpiecznego, wszakże będąc ? z łaski
3  Panskiej dobrze przed Urzędem w ? obrowolnie ze-
4  znawam. Jak gdziezby Pan Bog na miłe w tej niemocy mojej
5  smierc dopuscic raczył, ja rzeczy ruchome moje wszystkie ziemie mojej
6  od mała do wiela, zadnych nie wymieniac zapisuię, daruję i odkazuję
7  tym testamentem wiecznym darem nie wniesc prawa Przyia-
8  ciolom moim i krewnym nie zostawiając ktore rzeczy ona ....|
```



Current Limitations 2: No confidence subagent

There is no agent that independently about confidence, the same agent that runs OCR also gives a confidence score



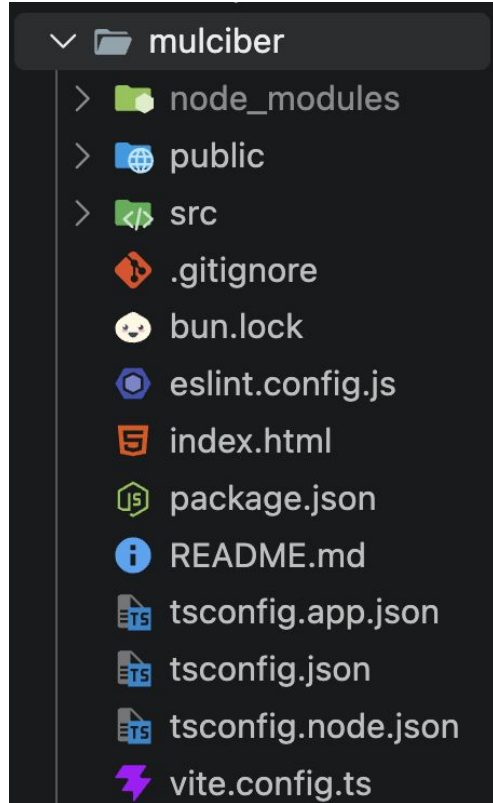
```
{
  "word": "Jacobi",
  "language": "Latin",
  "language_confidence_score": 1.0,
  "language_confidence_reasoning": "Latin given name in genitive.",
  "word_declension": null,
  "word_type": "name",
  "line_number": 1,
  "transcription_confidence_score": 0.95,
  "transcription_confidence_reasoning": "Clearly written."
},
{
  "word": "Szanc",
  "language": "Polish",
  "language_confidence_score": 0.95,
  "language_confidence_reasoning": "Polish surname.",
  "word_declension": "nominative singular",
  "word_type": "name",
  "line_number": 1,
  "transcription_confidence_score": 0.9,
  "transcription_confidence_reasoning": "Distinct spelling 'Szanc'."
},
}
```

Current Limitations 3: Inconsistent predicate

	subject_id	predicate	object_id
57	PER-0034	nupta_fuit	PER-0035
58	PER-0034	nupta_est	PER-0036
59	PER-0036	habitat	LOC-0011
14	PER-0013	member_of	ORG-0003
15	PER-0019	spouse_of	PER-0018
16	PER-0018	legare	PER-0019

Should be in
same category!

Other Files: multiciber/



Potential Future Frontend

